Bureau International des Poids et Mesures

Le système international d'unités

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Annexe 2

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Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2

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Ytterbium 171 Ion ()

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CCL-CCTF Frequency Standards Working Group

April 13, 2022
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Bureau International
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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Ytterbium 171 Ion ()

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1. Recommended value CCTF 22nd Meeting (2020) of the frequency in the CIPM List of Frequencies

$f\left({}^{171}\mathrm{Yb}^{+},\ \text{octupole}\right) = 642\ 121\ 496\ 772\ 645.12\ \mathrm{Hz}$

equivalent to

$\lambda\left({}^{171}\mathrm{Yb}^{+}\right) = 466\ 878\ 090.060\ 496\ 78\ \mathrm{fm}$

with a relative standard uncertainty of 1.9×10^{-16}.

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second CIPM REC 2 (2015, E).

2. Method to establish the recommended value

A global adjustment of all measurements of frequency ratios published in peer-reviewed publications and available to the CCL-CCTF WGFS was carried out following the methods presented in [3] to [7] .

This adjustment determines the frequency of 14 transitions (see Figure 1, see p. 5) which are either already adopted as secondary representations of the second [7] or considered as candidates for SRS. It took into account 105 measurements, including 33 frequency ratios and 72 absolute frequency measurements (i.e. ratios to the ^{133}Cs frequency). A total of 483 correlations between these input measurements were estimated and considered in the adjustment. More details on the input data and the processing are provided at https://webtai.bipm.org/ftp/pub/tai/publication/wgfs/Adjustment_2021.html . The recommended value is the direct result of the adjustment, rounded as deemed adequate with respect to the recommended uncertainty.

While the results are from a global adjustment, it is of interest to note (see Figure 1, see p. 5) that the ^{171}Yb^{+} transition is involved in 7 measurements relative to ^{133}Cs, and in 9 frequency ratios with optical transitions.

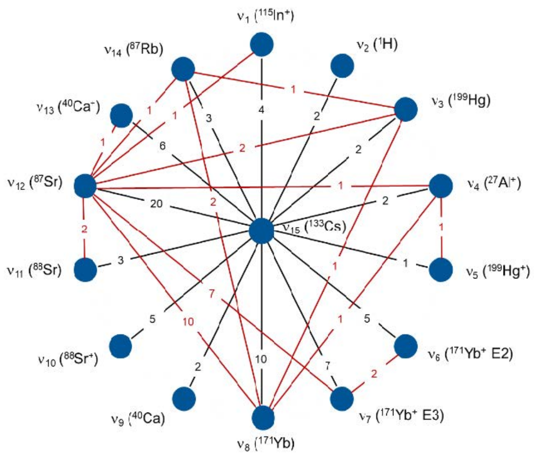

Figure 1 — Representation of the 105 measurements linking 14 transitions on the circle and ^{133}Cs at the center.

$^{171}Yb^{+}$ ion, $6s^{2}S_{1/2} - 4f^{13}s^{22}F_{7/2}$ unperturbed optical transition

References

[1] CCTF 22nd Meeting (2020) Consultative Committee for Time and Frequency (2020) *22nd meeting of the CCTF* (Paris: International Bureau of Weights and Measures) https://www.bipm.org/en/committees/cc/cctf/22-_1-2020.

[2] CIPM REC 2 (2015, E) International Committee for Weights and Measures (2015) *Updates to the list of standard frequencies* (Paris: International Bureau of Weights and Measures) https://www.bipm.org/en/committees/ci/cipm/104-_1-2015/resolution-2.

[3] H S Margolis and P Gill (2015) Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values. *Metrologia*. **52** (5) 628–634. https://doi.org/10.1088/0026-1394/52/5/628.

[4] L Robertsson (2016) On the evaluation of ultra-high-precision frequency ratio measurements: examining closed loops in a graph theory framework. *Metrologia*. **53** (6) 1272–1280. https://doi.org/10.1088/0026-1394/53/6/1272.

[5] G. Panfilo, communication to the CCL-CCTF WGFS. A new implementation of [4] was realized in MatLab at the BIPM (2020)

[6] Ch. Oates, communication to the CCL-CCTF WGFS. An independent program was developed in Mathematica at NIST (2017)

[7] Fritz Riehle, Patrick Gill, Felicitas Arias and Lennart Robertsson (2018) The CIPM list of recommended frequency standard values: guidelines and procedures. *Metrologia*. **55** (2) 188–200. https://doi.org/10.1088/1681-7575/aaa302.